

Kaaval Assurance

**Routers predict.
Kaaval verifies.**

An inference assurance plane for open-weight models on AMD compute.

Verify, not predict. Receipts, not promises.

AI INFRASTRUCTURE TOOL

ACCELERATE. VALIDATE. SCALE.



AI Decisions are Liabilities.

Unverified local models approve unauthorized refunds, create exposure, and fail silently.

Standard guardrails check syntax; they cannot verify business logic. As agents handle transactions, quality drift translates directly into financial risk.



Gate. Verify. Escalate.

Local open-weight inference checked by code contracts in <1ms. Only fails safe.

Kaaval intercepts response structures locally. If checks fail, it routes to a remote backup, saving **88%** of API costs while guaranteeing compliance.



3-Layer Assurance Engine

1. Conformance Gate

Deterministic JSON shape, enums, & grounding checks in <1ms.

2. Adaptive EWMA Routing

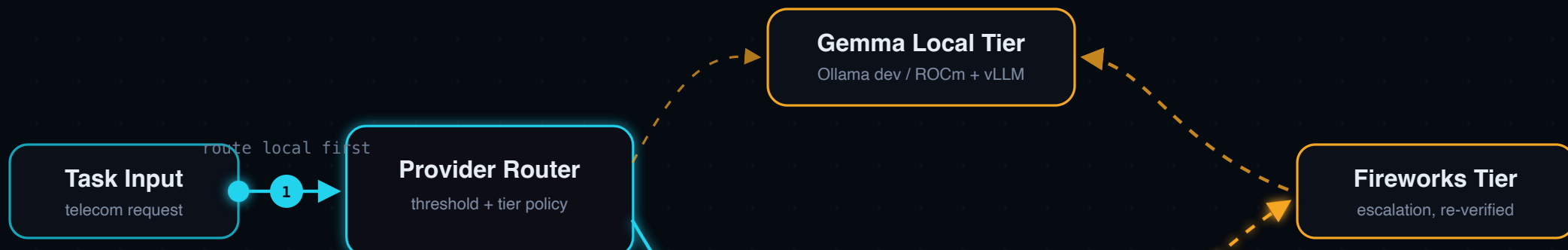
Tracks category failures in real-time. Pre-routes directly to remote on drift.

3. Sampled Offline Audit

Adversarial critic challenges accepted answers with FP-calibration.

INFERENCE ASSURANCE FLOW

local-first Gemma tier -> contract gate -> verified answer or governed escalation



Measured Proof on AMD Silicon

100%

[MEASURED]

Contract Conformance

88%

[MEASURED]

API Costs Saved

0%

[MEASURED]

Escalation Rate (Proof Bun)

Verify, Don't Predict.

Traditional Routers

Focus on **guessing** model quality *before* generation. Works for content writing, but fails when business contracts cannot tolerate errors.

Kaaval Assurance

Focus on **measuring** contract conformance *after* generation. Built for transactional, policy-bound AI decisions and automated compliance.